MILITARY SPECIFICATION

CARTRIDGE, 40MM, YELLOW SMOKE, GROUND MARKER, M716

LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, packaging, the fabrication, manufacture of one type of cartridge designated as Cartridge, 40mm, Yellow Smoke, Ground Marker, M716.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. - The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC: 1310

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1234 - Pyrotechnics Sampling, Inspection and Testing

DRAWINGS (See 6.7)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT AND PACKING DRAWINGS

9323265 - Cartridge, 40mm, Yellow Smoke, Ground Marker, M716
9209204 - Box, Ammo, Metal for Cartridge, 40mm, White Star, Cluster or Smoke Canopy or Parachute
9209205 - Box, Wirebound, Packing, Ammunition for Cartridge, 40mm, White Star, Cluster or Smoke Canopy or Parachute

INSPECTION EQUIPMENT DRAWINGS

9202770 - Chamber Gage
9202783 - Alignment Check

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

TECP 700-700 Volume 4 - TECOM Material Test Procedure

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402. Orders should specify "49 CFR 100-199 (latest revision)").

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with the applicable drawings and specifications.

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3.2 Cartridge - The cartridge shall comply with all requirements specified on Drawing (Dwg.) 9323265, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Transportation vibration. - The cartridge shall comply with the following requirements:

3.3.1 The cartridge shall be safe to transport following this test (See 6.5).

3.3.2 There shall be no evidence of external damage to the cartridge that will affect the intended function.

3.4 Functioning. - The cartridge shall comply with the following requirements;

a. Mean velocity of 247±7 feet per second and a standard deviation not exceeding 4.5 feet per second. (Major)

b. No projectile shall stick in the gun bore. (Critical)

c. Smoke duration time of usable visibility of 25 seconds, minimum. (Major)

d. Smoke ignition on impact or within 10 seconds of launch at hot and ambient phase and 15 seconds at cold phase. (Major)

e. Incorrect smoke color. (Critical)

f. Fuze must function. (Major)

g. No smoke ignition in gun bore. (Critical)

h. No flame duration of more than 3 seconds. (Major)

3.4.1 Hot temperature (160 degrees F. plus or minus 5 degrees F.). - The cartridge shall comply with the requirements of 3.4.

3.4.2 Ambient temperature (70 degrees F. plus or minus 5 degrees F.). - The cartridge shall comply with the requirements of 3.4.

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3.4.3 Cold temperature (-65 degrees F. plus or minus 5 degrees F.). - The cartridge shall comply with the requirements of 3.4.

3.5 First article inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship. - All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions . - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. - The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. - The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Projectile	9323266	25
Cartridge	9323265	465 (440 to a Government Proving Ground for Ballistic Testing)

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4.3.2 Inspections to be performed. - See MIL-A-48078 and Table I herein.

4.3.3 Rejection. - See MIL-A-48078 and the following:

4.3.3.1 Proving ground tests. -

a. Any critical defects occurs in any phase (see 3.4).

b. A total of eight (8) or more in the hot phase fail any of the following:

(1) Smoke duration time.
(2) Smoke ignition time (average).
(3) Fuze must function.
(4) Flame duration more than 3 seconds.

c. A total of twenty-two (22) or more in the ambient phase fail any of the following:

(1) Smoke duration time.
(2) Smoke ignition time (average).
(3) Fuze must function.
(4) Flame duration more than 3 seconds.

d. A total of eight (8) or more in the cold phase fail any of the following:

(1) Smoke duration time.
(2) Smoke ignition time (average).
(3) Fuze must function.
(4) Flame duration more than 3 seconds.

e. Mean velocity or standard deviation fail the applicable requirement (average).

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63131 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Cartridge and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile assembly (Dwg 9323266)				
	Examination for defects	25		3.2	4.4.2.1
	Static burn time	25		3.2	4.5.2
	Cartridge (Dwg. 9323265)				
	Examination for defects	25		3.2	4.4.2.2
	Pull test of projectile	25		3.2	4.5.3
	*Transportation-vibration	440		3.3	4.5.4
	*Air pressure	440		3.2	4.5.5
	Functioning				
	Hot	80		3.4.1	4.5.6.1
	Ambient	280		3.4.2	4.5.6.2
	Cold	80		3.4.3	4.5.6.3

*Prior to submission of the 440 samples to a Gov't proving ground for functioning tests, the samples shall be inspected and accepted at the contractors plant for these tests.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot shall contain:

a. Cartridge case loaded assemblies from one interfix lot number from one manufacturer.

b. Projectile assemblies from one interfix lot number from one manufacturer.

c. Fuze assemblies from one interfix lot number from one manufacturer.

d. Ignition mix from one interfix lot number from one manufacturer.

e. Smoke composition produced by one manufacturer under one continuous set of operating conditions and which consist of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

f. Delay assemblies from one interfix lot number from one manufacturer.

4.4.2 Examination. - See MIL-A-48078.

a. Sampling plans. - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63131 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Projectile assembly		9323256
			NEXT HIGHER ASSEMBLY 9323255

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Static burn time	32	1.5%	3.2	4.5.2
102	Weight		0.40%	3.2	Balance
103	Total length		0.40%	3.2	Gage
104	Tape missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63131 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Cartridge		9323265
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Chamber gage failure		100%	3.2	9202770 9202783 8827895
2	Marking incorrect		100%	3.2	Visual
Major					
101	Marking unidentifiable	50	0.40%	3.2	Visual
102	Pull test		2.50%	3.2	4.5.3
103	Transportation-vibration			3.3	4.4.3.3, 4.4.3.7 and 4.5.4
104	Air pressure			3.2	4.4.3.2, 4.4.3.4 4.4.3.8 and 4.5.5
105	Functioning			3.4	4.4.3.5, 4.4.3.9 and 4.5.6
Minor					
201	Total length		0.65%	3.2	Gage
	*Prior to forwarding the samples to a Gov't proving ground for functioning tests, the samples shall be inspected and accepted at the contractors plant for these tests.				

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63131 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Unsealed Metal Box		9209204
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Incorrect number of cartridges		0.40%	3.2	Visual
102	Desiccant bags missing		0.40%	3.2	Visual
103	Rubber gasket damaged or improperly assembled		0.40%	3.2	Visual
Minor					
201	Fillers missing		0.65%	3.2	Visual
202	Tube missing		0.65%	3.2	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63131 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9209204
4.4.2.4	Sealed Metal Box				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Air pressure	*	*	3.2	4.5.7
Minor					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual
202	Contents loose		0.65%	3.2	Manual
	*One (1) packed, sealed box shall be selected from each twenty (20) boxes packed. If one &1) box fails, the quantity represented by the sample shall be rejected.				

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63131 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Sealed Wooden Packing Box				9209205
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Minor					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual
202	Contents loose		0.65%	3.2	Manual
203	ICC nomenclature missing		0.65%	3.2	Visual

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4.4.3 Testing.

4.4.3.1 Moisture content (see Table I) Major. -

Table I

Material	Drawing
Composition, Ignition Smoke	9323280
Composition, Pyrotechnic, Yellow Smoke	9323276

At the time and place of loading, the contractor shall provide adequate controls to insure compliance with the requirements and shall test for verification at least one (1) sample of each material from each eight (8) of each hours production of cartridges. Composite samples shall not be used. If the moisture content of a sample exceeds the requirement and loading has not begun, that quantity of material represented by the sample shall be rejected. If assemblies have been loaded with material containing excessive moisture, the remaining material that is represented by the sample together with all cartridges loaded with the nonforming material shall be rejected.

4.4.3.2 Air pressure. - This test shall be conducted 100 percent. Any cartridge which fails to comply with the requirement specified on the applicable drawing shall be classified defective and removed from the lot.

4.4.3.3 Transportation-vibration. - After completion of the 100 percent air pressure test and beginning with the first lot produced and continuing until three (3) acceptable lots (lots not accepted by disposition) have complied with the acceptance criteria specified in TABLE II, two-hundred and sixty-four (264) cartridges shall be selected from each lot, packed in their regular shipping containers and subjected to this test. The cartridges shall be observed and examined visually with disassembly for any evidence of failure to comply with the requirements of TABLE II.

TABLE II

Defect	Drawing
Cartridge not safe to transport following test	Critical
Cartridge damaged after test	Major

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The lot shall be rejected if any defect as classified in TABLE II occurs.

4.4.3.4 Air pressure. - After completion of the transportation-vibration test, the two-hundred and sixty-four (264) samples from each lot shall be subjected to this test. If nine (9) or more cartridges fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. Cartridges which fail this test shall be discarded and replaced with units which have been successfully subjected to transportation-vibration test prior to subsequent testing.

4.4.3.5 Functioning. - After completion of the air pressure test and beginning with the first lot produced and continuing until three (3) acceptable lots (lots not accepted by disposition) have complied with the criteria of TABLE III, the two-hundred and sixty-four (264) samples from each lot shall be packed and shipped to a Government proving ground and tested as follows:

4.4.3.5.1 Hot. - Forty-four (44) samples shall be subjected to this test.

4.4.3.5.2 Ambient. - One-hundred and seventy-six (176) samples shall be subjected to this test.

4.4.3.5.3 Cold. - Forty-four (44) samples shall be subjected to this test.

4.4.3.5.4 Rejection. - The lot shall be rejected if:

a. Any critical defect occurs in any phase (see 3.4).

b. A total of six (6) or more in the hot phase fail any of the following:

(1) Smoke duration time.
(2) Smoke ignition time (average).
(3) Fuze must function.
(4) Flame duration more than 3 seconds.

c. A total of fifteen (15) or more in the ambient phase fail any of the following:

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(1) Smoke duration time.
(2) Smoke ignition time (average).
(3) Fuze must function.
(4) Flame duration more than 3 seconds.

d. A total of six (6) or more in the cold phase fail any of the following:

(1) Smoke duration time.
(2) Smoke ignition time (average).
(3) Fuze must function.
(4) Flame duration more than 3 seconds.

e. Mean velocity or standard deviation fail the applicable requirement (average).

4.4.3.6 Suitability test plan (informational test only). - Beginning with the first lot produced by each producer and continuing for three (3) accepted lots (lots not accepted by disposition) that are accepted in accordance with 4.4.3.4, an extra sample of eighty-eight (88) rounds shall be selected from each lot and preconditioned and tested as follows:

a. The eighty-eight (88) rounds shall be packaged in their regular shipping containers and subjected to transportation vibration environmental in accordance with MIL-STD-331, Test No. 104.

b. After transportation vibration, the rounds shall be unpacked and subjected to the JAN temperature and humidity cycle (14 days) specified in MIL-STD-331, Test No. 105.

c. After temperature and humidity cycle, the rounds shall be temperature conditioned and fired as follows:

30 rounds +165 degrees F. ±5 degrees F.
28 rounds ambient (between +40 and +110 degrees F.)
30 rounds -65 degrees F. ±5 degrees F.

d. Fire the rounds in accordance with paragraph 4.5.6.4 and with TECOM Material Test Procedure, TECP 700-700 Volume 4. Test results, failure analysis reports and other information as requested shall be forwarded to ARRADCOM, ATTN: DRDAR-QAA-Q.

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4.4.3.7 Transportation-vibration, regular production. - After three (3) acceptable lots have complied with 4.4.3.3 and 4.4.3.4, one-hundred and thirty-two (132) of the samples subjected to the 100% air pressure test shall be selected from each lot, packed in their regular shipping containers and subjected to this test. The lot shall be rejected if any defect as classified in TABLE II occurs.

4.4.3.8 Air pressure, regular production. - After completion of the transportation-vibration test, the one-hundred and thirty-two (132) samples shall be removed from their shipping containers and subjected to this test. If eight (8) or more samples fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. Cartridges which fail this test shall be discarded and replaced with units which have been successfully subjected to transportation-vibration test prior to subsequent testing.

4.4.3.9 Functioning, regular production. - After three (3) acceptable lots have complied with 4.4.3.5, the one-hundred and thirty-two (132) samples subjected to the air pressure test shall be packed and shipped to a Government proving ground and tested as follows:

4.4.3.9.1 Hot. - Twenty-six (26) samples shall be subjected to this test.

4.4.3.9.2 Ambient. - Eighty (80) samples shall be subjected to this test.

4.4.3.9.3 Cold. - Twenty-six (26) sample shall be subjected to this test.

4.4.3.9.4 Rejection. - The lot shall be rejected if:

a. Any critical defect occurs in any phase (see 3.4).

b. A total of four (4) or more in the hot phase fail any of the following:

(1) Smoke duration time.
(2) Smoke ignition time (average).
(3) Fuze must function.
(4) Flame duration more than 3 seconds.

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c. A total of eight (8) or more in the ambient phase fail any of the following:

(1) Smoke duration time.
(2) Smoke ignition time (average).
(3) Fuze must function.
(4) Flame duration more than 3 seconds.

d. A total of four (4) or more in the cold phase fail any of the following:

(1) Smoke duration time.
(2) Smoke ignition time (average).
(3) Fuze must function.
(4) Flame duration more than 3 seconds.

e. Mean velocity or standard deviation fail the applicable requirement (average).

4.4.4 Inspection equipment. - The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.

4.5.1.1 Ignition smoke composition, and pyrotechnic yellow smoke composition. - The moisture content of each of these compositions at the time and place of loading shall be determined in accordance with Method 102.1 of MIL-STD-1234 or any other approved method with the exception that the smoke compositions be dried for four hours at ambient temperature prior to testing.

4.5.2 Static burn time of projectile assembly. - This test shall be conducted in accordance with any Government approved procedure.

4.5.3 Pull test of cartridge. - The cartridge shall be placed in an approved fixture and the axial force specified on the applicable drawing shall be applied. Cartridge shall be pulled until total separation occurs and data shall be recorded. This test is a destructive test. Parts so tested shall not be returned to the lot.

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4.5.4 Transportation vibration. - The cartridges shall be packaged and packed in accordance with dwgs. 9209204 and 9209205 and tested in accordance with Test No. 104 of MIL-STD-331, except that each box shall be vibrated at the specified amplitudes for four (4) hours in each of three (3) different positions (i.e., box positioned so that the cartridges are vertical with case end down, box positioned so that the cartridges are horizontal, and box positioned so that the cartridges are vertical with case end up. After the test, the box packing and the cartridges shall be examined to determine compliance with the requirements. Cartridges shall be used for subsequent tests.

4.5.5 Air pressure. - The cartridge shall be placed in an approved fixture and a measured quantity of air shall be applied to produce the required air pressure. Cartridges shall be used for subsequent testing.

4.5.6 Functioning. - The following tests shall be performed at a Government owned proving ground:

4.5.6.1 Hot temperature. - The cartridges shall be packed in their regular shipping containers and conditioned for a minimum of 16 hours at a temperature of 160 degrees F. plus or minus 5 degrees F. The cartridges shall be taken from the conditioning chamber, removed from the packing container, and fired in accordance with the procedure specified in 4.5.6.4.

4.5.6.2 Ambient temperature. - The cartridges shall be packed in their regular shipping containers and conditioned for a minimum of 16 hours at a temperature of 70 degrees F. plus or minus 5 degrees F. The cartridges shall be taken from the conditioning chamber, removed from the packing container, and fired in accordance with the procedure specified in 4.5.6.4.

4.5.6.3 Cold temperature. - The cartridges shall be packed in their regular shipping containers and conditioned for a minimum of 16 hours at a temperature of -65 degrees F. plus or minus 5 degrees F. The cartridges shall be taken from the conditioning chamber, removed from the packing container, and fired in accordance with the procedure specified in 4.5.6.4.

4.5.6.4 Firing procedure. - Within fifteen (15) minutes after being removed from the temperature soak, the cartridge shall be fired from an approved launcher, placed on an approved mount to impact at 300 meters minimum from the

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launcher on a six inch thick loose gravel pad. The gravel used shall have a 1/2 inch maximum size. The velocity shall be measured at 20 feet from the muzzle of the launcher using two sets of screens, each set of screens having an electronic counter. One set of screen shall be placed 14 and 26 feet from the launcher and the other set 15 and 25 feet from the launcher. NOTE: To insure that the chronographs display accurate time intervals, the ratio of the two times shall not differ from 1.200 by more than plus or minus 0.005. If the ratio differs from 1.200 by more than plus or minus 0.005, the velocity results for that round shall be discarded and excluded from the calculation of mean velocity and velocity standard deviation. If the applicable number of improper ratios is exceeded, additional round(s) must be fired. Observation shall be made and results recorded for functioning, mean velocity and standard deviation (See 6.8).

4.5.6.4.1 Test validity. - If for any reason the proving ground considers the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

4.5.7 Air pressure of packed sealed metal box. - The sealed metal box shall be placed in an approved fixture and a measured quantity of air shall be applied to produce the required air pressure. Observation shall be made for leakage. This test is a non-destructive test. Boxes so tested may be returned to the lot.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level A. - The cartridges shall be preserved and packaged in accordance with dwg. 9209204.

5.2 Packing

5.2.1 Level A. - The cartridges, preserved and packaged as specified in 5.1.1, shall be packed in accordance with dwg. 9209205.

5.3 Marking. - Marking shall be in accordance with Dwgs. 9209204, 9209205, and Code of Federal Regulations, Title 49, Parts 100-199.

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5.4 Shipping. - When shipments from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. - This specification covers the loading, assembling and packing for Cartridge, 40mm, Yellow Smoke, Ground Marker, M716.

6.2 Ordering data. - See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. - See MIL-A-48078. Submit equipment designs as required to Commander, ATTN: DRDAR-QAA-I, ARRADCOM.

6.4 Submission of results of contractor-conducted examinations and tests. - Data shall be submitted in accordance with data item DI-R-1721 on the DD Form 1423 for the contract.

6.5 Submission of test data. - In addition to the normal distribution of records, when the cartridge is procured by the US Army Armament Command, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commander, ARRADCOM, Attn: DRDAR-QAA-Q.

6.6 The cartridges will be considered safe to transport providing no evidence exists of loose powder or composition in the box.

6.7 Proving ground test summary:

TEST	SAMPLE SIZE	REQUIREMENTS
Functioning		
(1) First Article		
a. Hot	80	See 3.4.1 & 4.3.3.1
b. Ambient	280	See 3.4.2 & 4.3.3.1
c. Cold	80	See 3.4.3 & 4.3.3.1
(2) First 3 lots		
a. Hot	44	See 3.4.1 & 4.4.3.5
b. Ambient	176	See 3.4.2 & 4.4.3.5
c. Cold	44	See 3.4.3 & 4.4.3.5

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(3)	Regular Production		
	a. Hot	26	See 3.4.1 & 4.4.3.9
	b. Ambient	80	See 3.4.2 & 4.4.3.9
	c. Cold	26	See 3.4.3 & 4.4.3.9

6.8 Drawings. - Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of ARRADCOM.

6.9 Standard deviation shall be calculated from the following formula or other approved formula:

$$S = \sqrt{\frac{\Sigma(X - \bar{X})^2}{n - 1}}$$

where:

X = each individual value

$\bar{X}$ = Sample arithmetic mean = $\frac{\Sigma X}{n}$

n = sample size

$\Sigma(X - \bar{X})^2$ = the sum of the squares of the differences between the sample mean and each individual value

$n - 1$ = one less than the sample

6.9.1 For a faster and easier method of arriving at the standard deviation, the following alternate method may be used.

$$S = \sqrt{\frac{n \Sigma x - (\Sigma X)^2}{n (n - 1)}}$$

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